

Supplementary Materials

A Numerical Stability

Optimizing complex likelihood expressions using gradient descent requires attention to numerical stability. The marginal likelihoods of many common click models contain products of small probabilities, which can lead to numerical underflow in finite-precision computer arithmetic [18, 28]. Below, we cover the techniques CLAX uses to stabilize complex likelihood expressions by performing all probability computations in log-space.

A.1 Multiplication

By moving to log-probabilities, products of probabilities simplify to sums (and division to subtraction):

$$\log \left(\prod_{i=1}^n p_i \right) = \sum_{i=1}^n \log p_i, \quad (12)$$

which essentially eliminates the concern of numerical underflow when multiplying small probabilities.

A.2 Addition

While multiplication becomes stable (and faster) in log-space, the addition of probabilities becomes more complicated as it requires first exponentiating log probabilities. This reintroduces the problems we seek to avoid, as exponentiating large positive inputs lead to overflow and exponentiating large negative inputs lead to underflow. The standard solution is to avoid large inputs to the $\exp(\cdot)$ operation via the log-sum-exp trick [1]:

$$\log_sum_exp(a) = a_{\max} + \log \left(\sum_{i=1}^n \exp(a_i - a_{\max}) \right), \quad (13)$$

where $a = (a_1, \dots, a_n)$ is a vector of log values and $a_{\max} = \max_i(a_i)$ is the maximum input value. The trick is prevalent in probabilistic modeling, and we also use it to transform the output logits of neural networks $x \in \mathbb{R}$ to log-probabilities by implementing numerically stable versions of the log-sigmoid functions:

$$\begin{aligned} \log(\sigma(x)) &= -\log_sum_exp([0, -x]), \text{ or} \\ \log(1 - \sigma(x)) &= -\log_sum_exp([0, x]). \end{aligned} \quad (14)$$

A.3 Complements and cancellation

Sometimes we need to compute the log of a complement $\log(1 - p)$, e.g., in the binary-cross entropy loss or when computing log-posteriors in the DBN [5]. Performing this step directly from log-probability $\log p$ requires computing: $\log(1 - \exp(\log p))$. This expression is numerically unstable in two ways: (i) underflow: when p is very small, $\log p$ is very negative, causing $\exp(\log p)$ to underflow to zero; and (ii) catastrophic cancellation: when $p \approx 1$, we have $\exp(\log p) \approx 1$, making $1 - \exp(\log p) \approx 0$, since subtracting nearly equal floating point numbers leads to a loss of precision [18]. Therefore, we compute $\log 1mexp(x)$ as proposed in [35] and adopted by major frameworks such as TensorFlow¹¹ and JAX.¹²

Mächler [35] proposes a piecewise approximation that switches between two stable expressions that are precise in different input

ranges.¹³ For a log-probability $a \in \mathbb{R}$, $a \leq 0$:

$$\log 1mexp(a) = \begin{cases} \log(-\expm1(a)) & \text{if } a > -\log(2) \\ \log 1p(-\exp(a)) & \text{if } a \leq -\log(2). \end{cases} \quad (15)$$

The implementation relies on the standard functions $\log 1p(x)$, which accurately computes $\log(1 + x)$, and $\expm1(x)$, which accurately computes $\exp(x) - 1$, to avoid catastrophic cancellation.

To summarize, CLAX performs all probability computations in log space for increased numerical stability, avoiding underflow and overflow as well as catastrophic cancellation. We list all implemented log-likelihoods in Appendix B and their corresponding implementation can be found in our code repository.¹⁴

B Models

We introduce the ten click models implemented in CLAX following the standard work of Chuklin et al. [9, Table 3.1] and their click log-probabilities. If we do not explicitly state a conditional click probability, unconditional and conditional click probabilities are equal for the specific model.

B.1 Global CTR model (GCTR)

The global CTR (GCTR) model, sometimes called the random click model, predicts a single average CTR across all documents. It serves as a simple baseline that any click model should surpass:

$$\log P(C = 1 \mid d, k) = \log p. \quad (16)$$

B.2 Rank-based CTR model (RCTR)

The rank-based CTR (RCTR) model assumes that click probability depends only on the rank k of a document and not its content. The model predicts the average CTR for all documents displayed at the same rank, treating them as equally attractive:

$$\log P(C = 1 \mid d, k) = \log \theta_k. \quad (17)$$

B.3 Document-based CTR model (DCTR)

The document-based CTR (DCTR) model assumes that clicks depend solely on a document and not on its position in the ranking:

$$\log P(C = 1 \mid d, k) = \log \gamma_d. \quad (18)$$

B.4 Position-based model (PBM)

The PBM assumes that clicks occurs only if a user first examines the result at rank k (with probability θ_k), and if the displayed document is attractive (γ_d):

$$\log P(C = 1 \mid d, k) = \log \theta_k + \log \gamma_d. \quad (19)$$

B.5 Cascade model (CM)

The cascade model (CM) assumes that users scan results from top to bottom, click on the first attractive document they find, and then immediately stop their search. The probability of a click at rank k depends on the displayed document d being attractive (γ_d) and all preceding documents being unattractive:

$$\log P(C = 1 \mid d, k) = \log \gamma_d + \sum_{i=1}^{k-1} \log(1 - \gamma_{d_i}). \quad (20)$$

¹¹ https://www.tensorflow.org/probability/api_docs/python/tfp/math/log1mexp

¹² https://docs.jax.dev/en/latest/_autosummary/jax.nn.log1mexp.html

¹³ Interested readers can find the motivation behind the switching point $\log(2)$ under [35, Section 2]. ¹⁴ <https://github.com/philippager/clax>

Note that the cascade model can only explain a single click per list. All other documents after the first click, by definition, have a click probability of 0. To avoid a log-likelihood of $-\infty$ in our conditional click predictions, we follow the common practice to assign a very small default click probability to all documents following a click [9]:

$$\log P(C = 1 \mid d, k, c_{<k}) = \begin{cases} \log \gamma_d, & \text{if } \sum_{i=1}^{k-1} c_i = 0 \\ \min_log_prob, & \text{otherwise.} \end{cases} \quad (21)$$

B.6 User browsing model (UBM)

The user browsing model (UBM) extends the PBM by assuming that the probability of examination at position k depends also on the position of the last clicked document k' . This is most easily demonstrated in the conditional log-probability of click:

$$\log P(C = 1 \mid d, k, c_{<k}) = \log \theta_{k,k'} + \log \gamma_d, \quad (22)$$

where k is the position of the current document and k' the position of the previously last clicked document. While conditional click probabilities are very simple, predicting clicks on a new list of documents is harder under the UBM, since it requires marginalizing over all possible last click positions $i < k$ before our current position:

$$\log P(C = 1 \mid d, k) = \log \left(\sum_{i=0}^{k-1} P(C = 1 \mid d_i, i) \cdot \left(\prod_{j=i+1}^{k-1} (1 - \theta_{j,i} \gamma_{d_j}) \right) \theta_{k,i} \gamma_d \right). \quad (23)$$

Each term in the sum represents a path to the current document: the probability of clicking at a previous rank i , then not clicking on anything until rank k , and finally examining and clicking the document at rank k given i was the last clicked position.

B.7 Dependent click model (DCM)

The dependent click model (DCM) is an extension of the cascade model to explain multiple clicks in a single ranking. The DCM assumes that users examine a list from top to bottom, click on relevant items, and after clicking have a rank-dependent probability λ_k to continue browsing:

$$\begin{aligned} \log P(C = 1 \mid d, k) &= \log(\epsilon_k) + \log(\gamma_d) \\ \log(\epsilon_{k+1}) &= \log(\epsilon_k) + \log(\gamma_{d_k} \lambda_k + (1 - \gamma_{d_k})). \end{aligned} \quad (24)$$

When conditioning on observed clicks, the examination probability changes based on the actions in the current session:

$$\begin{aligned} \log P(C = 1 \mid d, k, c_{<k}) &= \log(\epsilon_k) + \log(\gamma_d) \\ \log(\epsilon_{k+1}) &= \log \left(c_k \lambda_k + (1 - c_k) \frac{(1 - \gamma_{d_k}) \epsilon_k}{1 - \gamma_{d_k} \epsilon_k} \right). \end{aligned} \quad (25)$$

If a user clicks on a document, they continue to the next rank with probability λ_k and if they do not click, we calculate the posterior probability of examining the next rank given that we observed no click using Bayes' rule.

B.8 Click chain model (CCM)

The click chain model (CCM) is an extension of the DCM, assuming a total of three continuation scenarios that do not only explain continuation after clicking a document but also allow users to abandon

a session without any clicks. First, τ_1 is the probability of a user continuing to the next document after not clicking on the current document. Second, if the user clicks on the current document but is not satisfied, τ_2 is the probability of the user continuing to the next position. And lastly, τ_3 is the probability that a user clicks on the current item, finds it satisfying, but still wants to continue to the next document:

$$\begin{aligned} \log P(C = 1 \mid d, k) &= \log(\gamma_d) + \log(\epsilon_k) \\ \log(\epsilon_{k+1}) &= \log(\epsilon_k) \\ &\quad + \log(\gamma_{d_k} ((1 - \gamma_{d_k}) \tau_2 + \gamma_{d_k} \tau_3) \\ &\quad + (1 - \gamma_{d_k}) \tau_1). \end{aligned} \quad (26)$$

When conditioning on the observed clicks, the update rule for the examination probability changes based on the user's action at the current rank. If a click occurred, we compute continuation based on satisfaction (equal to attractiveness γ_d) and the continuation probabilities τ_2 and τ_3 . If no click was observed, we compute the posterior log probability of continuing to the next rank:

$$\begin{aligned} \log P(C = 1 \mid d, k, c_{<k}) &= \log(\gamma_d) + \log(\epsilon_k) \\ \log(\epsilon_{k+1}) &= c_k \left[\log(\gamma_{d_k} \tau_3 + (1 - \gamma_{d_k}) \tau_2) \right] \\ &\quad + (1 - c_k) \left[\log(1 - \gamma_{d_k}) + \log(\epsilon_k) \right] \\ &\quad + \log(\tau_1) - \log(1 - \gamma_{d_k} \epsilon_k). \end{aligned} \quad (27)$$

B.9 Dynamic Bayesian network (DBN)

The dynamic Bayesian network (DBN) model of [5] separates the concepts of a document being attractive (γ_d) and being satisfying (σ_d). A user stops their search only if they click on an attractive document and are satisfied by it. If they do not click or are not satisfied by the clicked document, they continue browsing with a global continuation probability λ :

$$\begin{aligned} \log P(C = 1 \mid d, k) &= \log(\gamma_d) + \log(\epsilon_k) \\ \log(\epsilon_{k+1}) &= \log(\epsilon_k) + \log(\lambda) + \log(1 - \gamma_{d_k} \sigma_{d_k}). \end{aligned} \quad (28)$$

The conditional click probability again takes the user's actions in the current session into account. If a click was observed, we compute the probability of continuation based on satisfaction. If no click was observed, we compute the posterior probability of continuing to the next item:

$$\begin{aligned} \log P(C = 1 \mid d, k, c_{<k}) &= \log(\gamma_d) + \log(\epsilon_k) \\ \log(\epsilon_{k+1}) &= \log(\lambda) + c_k \left[\log(1 - \sigma_{d_k}) \right] \\ &\quad + (1 - c_k) \left[\log(1 - \gamma_{d_k}) + \log(\epsilon_k) \right] \\ &\quad - \log(1 - \gamma_{d_k} \epsilon_k). \end{aligned} \quad (29)$$

C Proof of PBM equality

Proof that the gradient of the expected complete-data log-likelihood (Q -function), evaluated at the current parameter estimates, is equal to the gradient of the marginal log-likelihood [42] that is being used during gradient-based optimization:

$$\begin{aligned}
\frac{\partial Q(\theta_k, \gamma_d \mid \theta_k, \gamma_d)}{\partial \gamma_d} &= \sum_{(d', k, c) \in \mathcal{D}, d'=d} \frac{\hat{a}}{\gamma_d} - \frac{1 - \hat{a}}{1 - \gamma_d} \\
&= \sum_{(d', k, c) \in \mathcal{D}, d'=d} \frac{1}{\gamma_d} \left[c + (1 - c) \frac{(1 - \theta_k) \gamma_d}{1 - \theta_k \gamma_d} \right] - \\
&\quad \frac{1}{1 - \gamma_d} \left[1 - \left(c + (1 - c) \frac{(1 - \theta_k) \gamma_d}{1 - \theta_k \gamma_d} \right) \right] \quad (30) \\
&= \sum_{(d', k, c) \in \mathcal{D}, d'=d} \frac{c}{\gamma_d} - \frac{(1 - c) \theta_k}{1 - \theta_k \gamma_d} = \frac{\partial \mathcal{L}}{\partial \gamma_d}.
\end{aligned}$$